

Low temperature recovery of OFF-state stress induced degradation of AlGaIn/GaN high electron mobility transistors

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ABSTRACT

Thermal annealing is a widely used strategy to enhance semiconductor device performance. However, the process is complex for multi-material multi-layered semiconductor devices, where thermoelastic stresses from lattice constant and thermal expansion coefficient mismatch may create more defects than those annealed. We propose an alternate low temperature annealing technique, which utilizes the electron wind force (EWF) induced by small duty cycle high density pulsed current. To demonstrate its effectiveness, we intentionally degrade AlGaIn/GaN high electron mobility transistors (HEMTs) with accelerated OFF-state stressing to increase ON-resistance $\sim 182.08\%$ and reduce drain saturation current $\sim 85.82\%$ of pristine condition at a gate voltage of 0 V. We then performed the EWF annealing to recover the corresponding values back to $\sim 122.21\%$ and $\sim 93.10\%$, respectively. The peak transconductance, degraded to $\sim 76.58\%$ of pristine at the drain voltage of 3 V, was also recovered back to $\sim 92.38\%$. This recovery of previously degraded transport properties is attributed to approximately 80% recovery of carrier mobility, which occurs during EWF annealing. We performed synchrotron differential aperture x-ray microscopy measurements to correlate these annealing effects with the lattice structural changes. We found a reduction of lattice plane spacing of (001) planes and stress within the GaN layer under the gate region after EWF annealing, suggesting a corresponding decrease in defect density. Application of this low-temperature annealing technique for *in-operando* recovery of degraded electronic devices is discussed.

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Gallium nitride (GaN) based electronic devices are promising due to their superior figure of merit and efficiency over current Si-based technology. However, despite exceptional electrical performance, GaN devices have yet to reach full potential and commercialization over concerns about electrical and structural reliability. AlGaIn/GaN high electron mobility transistors (HEMTs) that manifest a sustained two-dimensional electron gas (2DEG), which provides exceptional mobility,^{1,2} are one such attractive GaN-based device. However, they also suffer from residual and *in-operando* stress buildup due to lattice constant and thermal expansion coefficient mismatch of the epitaxial layers, in addition to inverse piezoelectric stresses that develop under high electric field.³ The compounding effects of multiple contributions to stress trigger the nucleation of electrically active crystallographic defects that cause gradual degradation of electrical properties and even catastrophic failure.³

Electrical reliability testing of HEMTs is most often performed in the OFF-state,^{3–16} semi-ON state,^{17–20} or ON-state^{5,6,10,14,16,21} configurations, each of which exhibits different degradation mechanisms. In the OFF-state, the reverse bias induced high electric field causes expansion of AlGaIn by the inverse piezoelectric effect resulting in strain relaxation and lattice defects.¹⁹ Also, the carriers collide with each other during high voltage operation generating energetic hot electrons, which get injected toward the barrier and passivation layer and get trapped creating interface states or bulk traps.²² The semi-ON state degradation of GaN HEMTs is mostly caused by hot electrons.¹⁷ ON-state degradation includes hot electrons and thermally activated trapping of electrons in the passivation layer and the semiconductor epilayers creating lattice defects.^{10,23} Electrical degradation mechanisms and defect formation in different operating conditions and how to avoid such degradation by modifying the epitaxial structure and/or